# Shared Metadata Service

## Persistent Catalog Cache, Language-Service Seam, and STS2 Integration

Design, implementation, and validation review

For vscode-mssql PR #22836

**Central question.** How should one shared, key-correct metadata substrate turn SQL catalog rows into immutable, freshness-aware snapshots that can be reused by future language, AI, Object Explorer, Query Studio, and designer features—without putting network I/O on an interactive hot path, cross-publishing one database under another key, losing truth under multi-window cache races, or confusing a generic STS2 query lane with a dedicated metadata endpoint?

**Review outcome at the pinned head.** The architecture is directionally strong and the focused implementation is substantial, but six correctness gaps remain merge-blocking: offline network enforcement, fail-closed database identity plus session recycling, auxiliary-session serialization, cache-key binding, cross-window cache publication ordering, and duplicate serverless wake ownership.

| | |
|---|---|
| Prepared for: | Karl Burtram |
| Primary change: | `dev/karlb/query_04_metadata_service` at `9955ff9e6` |
| PR base: | `69f54be2a` on `main` |
| Related vscode ref: | `dev/query` at `5be148b1e` |
| STS2 reference: | `sqltoolsservice dev/query` at `fe8300939` |
| Refactor notes: | `kburtram/notes main` at `a548c0b84` |
| Report snapshot: | 28 August 2026, America/Los_Angeles |

This document combines an as-built design explanation with an evidence-qualified implementation review. A companion Markdown artifact contains exact file/line comments, suggested patches, and regression tests suitable for direct PR review or Codex execution.

# Contents

## How to read this report

This report distinguishes five kinds of statement so that design intent, current code, and validation evidence are not accidentally merged:

**Implemented**
: Present in the pinned PR head and traced to its implementation.

**Target invariant**
: A safety or behavior contract the substrate is intended to enforce.

**Validated**
: Exercised by a named test, protocol scenario, build lane, or independently observed repository state.

**Reviewed gap**
: A current-head mismatch between the target invariant and the implementation.

**Future contract**
: A downstream consumer or STS2 endpoint shape described by the refactor plans, but not shipped by PR #22836.

**Key takeaway**

PR #22836 supplies a reusable *metadata substrate*: catalog representation, live hydration, store/lease lifecycle, freshness policies, drift detection, optional persistence, preview composition, and observability. It deliberately does **not** replace the current language service, Object Explorer, or query experience. Today it reaches SQL Server through the existing STS2 connection/query protocol. A future dedicated STS2 metadata service is an adapter substitution, not a fact already delivered by this PR.

**Evidence boundary**

The source review is pinned to exact commits and current GitHub state. This report combines two evidence sets: independent local builds/tests already run at the pinned head, and a later static cross-repository review of the implementation and protocol contracts. GitHub Actions facts, local results, PR-author claims, review-machine measurements, and proposed-but-not-yet-run regressions are labeled separately rather than presented as one undifferentiated proof.

# 1 Review snapshot and evidence model

Table 1: Pinned sources used to reconstruct the feature and its surrounding refactor.

| Source | Pinned state | Use in this report |
|---|---|---|
| PR #22836 | head `9955ff9e6`; base `69f54be2a` | All 46 changed filenames, current discussions, Actions state, and exact line anchors. |
| `vscode-mssql dev/query` | `5be148b1e` | Related query/data-plane refactor history and already-merged platform seams. |
| `vscode-mssql dev/refactor` | ref absent at review time | No assumptions are made from a deleted branch; surviving notes and current/merged code are used instead. |
| `sqltoolsservice dev/query` | `fe8300939` | STS2 generated contract, invariants, runtime behavior, and executable one-active-query scenario. |
| `kburtram/notes main` | `a548c0b84` | Metadata substrate/cache/drift, language-service, STS2, and observability design intent. |

Evidence is ranked as follows:

1. **Pinned source and executable protocol contracts.** These establish control flow, ownership, serialization, identity, and persistence behavior.
2. **Independent local validation.** The extension, embedded perftest, sqltoolsservice solution, STS2, authentication, language, and ServiceLayer lanes were run locally; exact results and exclusions appear in table 10.
3. **Current GitHub Actions logs.** At the reviewed head, analysis, packaging, normalization, and platform launch checks are green; the aggregate build-and-test check is red. The unit runner reports 3,975 passing, 0 failing, and 12 skipped; the aggregate failure comes from SQL Projects unit exit code 1 and one of 17 VSIX smoke tests.
4. **PR-body and prior-review results.** Additional focused pass counts, packaging measurements, and review-machine timing probes are useful reported evidence and remain identified as such.
5. **Missing regression proof.** A safety claim is not treated as validated when the current tests avoid the relevant branch—for example, an offline test with polling disabled and a guaranteed cache hit.

**Current review disposition**

The design can remain the basis of the refactor, but the current head should receive changes before merge. The first six findings in section 16.4 affect truth, isolation, or retry ownership rather than optional polish. Additional hardening findings are documented for the same PR or an explicitly scheduled follow-up.

# 2 Executive design summary

The feature addresses a real architectural problem. Without a common substrate, each metadata-backed surface tends to grow its own connection, query set, cache, name-resolution behavior, freshness policy, and failure semantics. That duplication is expensive, but the larger risk is semantic disagreement: completion may believe a column exists while Object Explorer, AI context, or scripting sees a different generation or silently treats unavailable data as an empty catalog.

PR #22836 establishes one extension-host registry with two scopes:

- a **server catalog** keyed by a non-reversible server fingerprint; and
- a **database catalog** keyed by that fingerprint plus the requested database spelling.

Consumers acquire ref-counted leases. A database lease exposes a current immutable snapshot, deterministic name/search/schema-context APIs, lazy module definitions, DDL notification, refresh, auxiliary sections, and policy-routed freshness. The mutable builder, filesystem, credential store, and VS Code configuration remain behind composition boundaries.

The live engine uses the SQL Data Plane to run a progressive H0–H7 catalog ladder. Rows are accumulated in a compact structure-of-arrays builder and published atomically as a `CatalogSnapshot` with a process-local generation and explicit per-section readiness. The persistent cache serializes the same model into canonical, gzip-compressed, content-addressed payloads with a manifest, bounds, and privacy policy intersection.

Table 2: Executive scorecard: design target versus the pinned implementation.

| Area | Target contract | Current-head assessment |
|---|---|---|
| Model and publication | Immutable, deterministic generations with per-section truth. | **Implemented** Strong. Structure-of-arrays, deterministic projections, partial readiness, and scale fixes are well covered. |
| Shared lifecycle | One entry per key, ref-counted leases, bounded idle retention, complete disposal. | **Implemented** Strong after same-key single-flight and disposal fixes. |
| Database isolation | A requested key can publish only metadata proven to belong to that database; recovery must reopen a fresh session. | **Reviewed gap** Initial mismatch logs and continues, and the wrapper drops `recycle()`. |
| Freshness/offline | Consumers choose stale, validated, live, or network-forbidden behavior. | **Reviewed gap** Policy vocabulary exists, but store-level offline mode does not prevent backend resolution, cache-miss hydration, polling, refresh, or auxiliary work. |
| STS2 query ownership | One active query per physical session, including lazy auxiliary sections. | **Reviewed gap** Main hydration is serialized; different auxiliary sections can overlap on one cached session and receive `Sts2.Busy`. |
| Cache integrity | A payload is bound to its key, logical hash, and a globally ordered publication decision. | **Reviewed gap** Byte/shape defenses are strong, but key binding, logical-hash verification, and atomic cross-window authority remain incomplete. |
| Data-plane retry policy | Serverless wake/retry has one owner per open attempt. | **Reviewed gap** Top-level and public service-view paths both wrap the same open. |
| Consumer cutover | Language/AI/OE/Query Studio pin snapshots and degrade honestly. | **Future contract** No ordinary product surface is routed to the substrate in this PR. |

**Design invariant**

The load-bearing target is **honest degradation**: missing, failed, partial, stale, offline, identity-drifted, or cache-invalid metadata must remain distinguishable all the way to the consumer. An empty list, an old generation, or a successful event must never be used as a convenient substitute for "not known."

**Why the current gaps are blocking**

The six P1 findings can cause live network activity under an offline promise, metadata from the wrong database to be published under a requested key, valid auxiliary requests to fail nondeterministically, a copied cache entry to cross keys, an older window to become authoritative after a newer capture, or one serverless open to execute nested retry loops. These are correctness and isolation failures at the substrate boundary, so downstream consumers cannot safely compensate for them.

# 3 Scope and position in the refactor train

## 3.1 What lands in this PR

The 46-file change spans the model, live engine, shared store, persistent cache, host/data-plane wiring, localization, observability, focused tests, and the embedded perftest registry. Its center of gravity is five mechanisms:

1. **Catalog model.** Compact immutable snapshots, collation-aware resolution, deterministic projections, exact column/FK detail, readiness, and statistics.
2. **Live metadata engine.** Progressive hydration, dedicated sessions, lazy definitions, validation, DDL sniffing, digest polling, identity drift, focus/serverless governance, and retry containment.
3. **Shared store.** Server/database keys, single-flight first acquisition, ref-counted leases, bounded idle retention, auxiliary catalogs, and one extension-host composition root.
4. **Persistent cache.** Canonical codec, manifest, filesystem protocol, coordinator, policy intersection, corruption handling, multi-window behavior, and age/byte eviction.
5. **Host integration.** Private-preview gates, settings/commands/localization, SQL Data Plane hardening, shutdown flush, observability registration, and a perftest warm-acquire probe.

## 3.2 What does not land

The PR does not route completion, hover, diagnostics, scripting, Query Studio, Object Explorer v2, or AI context through the new store. It also does not add a native TypeScript language engine, dedicated metadata RPCs to STS2, replay bundles, or central diagnostic upload. Those are downstream consumers or platform increments.

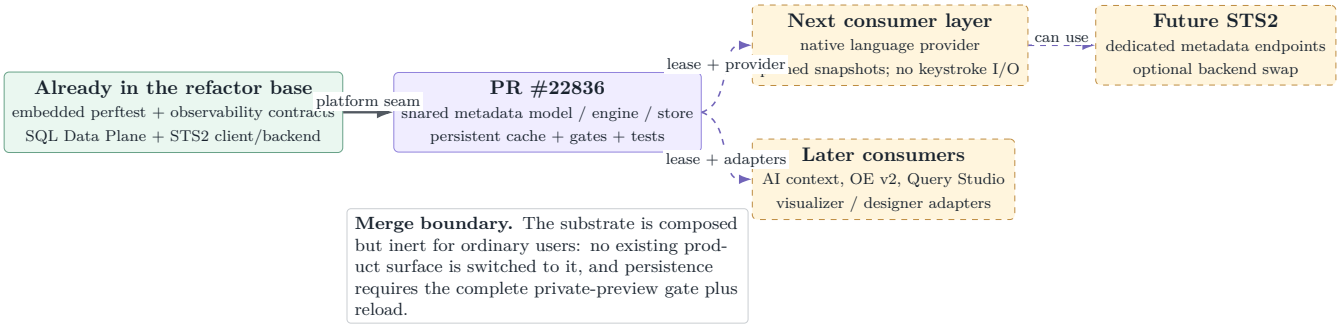



Figure 1: The PR is an enabling layer in the refactor train, not the consumer cutover. Dashed arrows are downstream contracts rather than current product routing.

# 4 Runtime architecture and composition

## 4.1 One extension-host composition root

`MainController.initializeMetadataStore()` configures host facts and storage before the first store request. `MetadataStoreService` lazily creates one `MetadataStore` per extension host. The metadata engine itself imports neither VS Code nor the filesystem; focus, poll cadence, data-plane enablement, global-storage path, and settings are injected.

The hierarchy of gates is strict:

$$\text{cache composed} \iff \texttt{enableExperimentalFeatures} \wedge \texttt{sqlDataPlane.enabled} \wedge \texttt{metadataCache.enabled}.$$

All three settings default off where applicable, and changing the dependency path requires a window reload. SQL Data Plane launch arguments also use the private-preview predicate, so merely setting the lower-level flag without the experimental umbrella does not activate STS2.

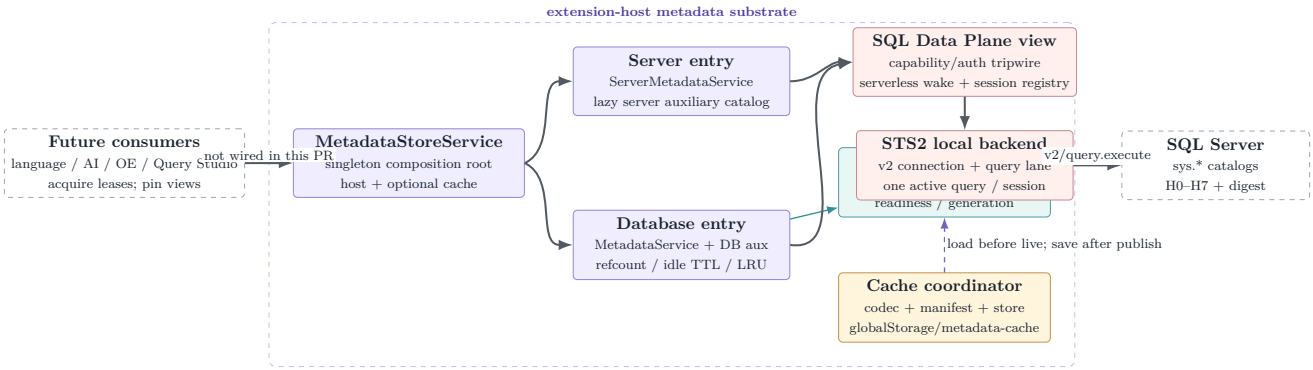



Figure 2: As-built runtime layering. The current transport is generic STS2 query execution; the cache is optional; consumer arrows remain a future integration surface.

## 4.2 Acquisition and the network-authorization boundary

The acquisition path has two logically separate responsibilities: recover the best local snapshot and, only when policy permits, create live infrastructure. The target ordering is cache-first and network-gated. This matters even when the backend does not open a physical SQL connection immediately: backend startup, polling, strict freshness, and auxiliary calls are all forms of live work that an explicit offline mode must prohibit.

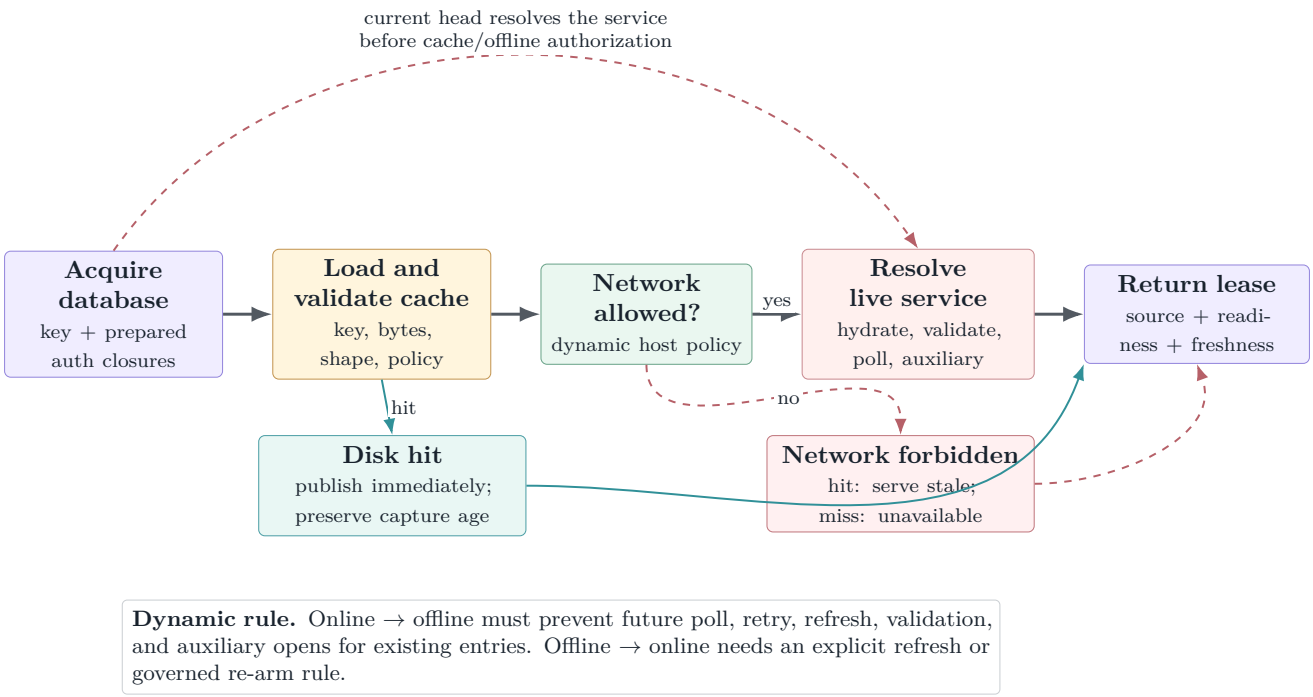



Figure 3: Target acquisition ordering and the current bypass. Offline is a network authorization boundary, not merely "do not schedule the first background refresh."

**Current-head review finding**

At `metadataStore.ts:417`, the current head resolves the SQL Data Plane service before cache load and before the offline outcome is known. On a miss, `MetadataService.acquire()` immediately hydrates; on a hit it still starts polling unless the test/config happens to disable it. Explicit refresh, strict freshness, and auxiliary sections are also callable. The existing offline test seeds a guaranteed hit and uses `pollSeconds=0`, so it does not prove the advertised contract.

### 4.3 Activation, maintenance, and shutdown

Composition is deliberately cheap. Cache maintenance is scheduled five seconds after activation and therefore does not contaminate `mssql.activate` timing. Debounced cache writes are flushed during extension deactivation before store disposal. Status commands are passive views of already-composed state; the SQL Data Plane status command is designed not to construct a backend or resolve credentials.

The cache-specific commands are:

- `mssql.metadataCache.showStatus`;
- `mssql.metadataCache.clearAll`;
- `mssql.metadataCache.clearForConnection`; and
- explicit enable/disable commands for offline mode.

## 5 Identity, authentication, and lease lifecycle

### 5.1 Two non-reversible fingerprints

The store never uses a password, token, or connection string as identity. It JSON-encodes connection-affecting tuples and hashes them with SHA-256, returning a 22-character base64url prefix:

- `pfp_*`: profile/session identity, including the default database;
- `sfp_*`: server identity, excluding the database.

JSON tuple encoding matters: delimiter-joined fields allow distinct legal values such as `["a|b","c"]` and `["a","b|c"]` to collide. The current PR fixes and tests this class for both fingerprints and derived profile IDs.

The database store key is `serverFingerprint + requested database spelling`. Retaining the spelling is reasonable before the catalog comparison rule is known, but it creates an obligation: the live session must prove that the database it opened is the database represented by the key before any snapshot or cache write is published.

### 5.2 Secrets stay deferred

`prepareConnection()` produces a sanitized profile reference and closures for authentication:

- SQL passwords are read from the credential store only when a physical session open requires them;
- Entra tokens are acquired only at physical open;
- integrated authentication carries no secret provider.

The SQL Data Plane registry evaluates declared requirements before invoking credential closures. That ordering is an important tripwire. However, the public service view and the top-level `openSession()` currently both own Azure serverless wake wrapping, producing a nested retry path. Capability checking, credential deferral, provider availability, and wake/retry ownership should remain separate concerns with one owner each.

### 5.3 Single-flight entries and ref-counted leases

First acquisition is single-flight per key for both server and database entries. Concurrent same-key callers join one creation rather than producing orphan engines or sessions. Each lease increments a refcount; disposal decrements it idempotently. A zero-ref database remains warm for a bounded idle period, and the least recently released idle entry is disposed first when the cap is exceeded.

A database entry owns:

- one session source for serialized hydration, digest validation, and lazy module definitions;
- one separate source for database auxiliary sections;
- one metadata engine, one lease handle, listeners, and cache-source state.

The resource split keeps ordinary metadata work away from user query sessions. It does *not*, by itself, serialize different auxiliary sections: the auxiliary engine coalesces callers for the same section but allows two different section keys to open the same cached session concurrently. STS2 documents and tests one active query per connection; the second execute returns `Sts2.Busy`. A catalog-wide auxiliary lane is therefore required.

### 5.4 Fail-closed database identity and recovery

The target identity protocol has two stages. The returned session metadata is a preliminary signal; H0's `DB_NAME()` plus `CATALOG_DEFAULT` comparison behavior is the authoritative proof. No H1–H7 rows should publish until that proof succeeds. A timeout or later rename drift must fence the current operation epoch, recycle the physical source, and reopen by the requested database name before clearing the drift latch.

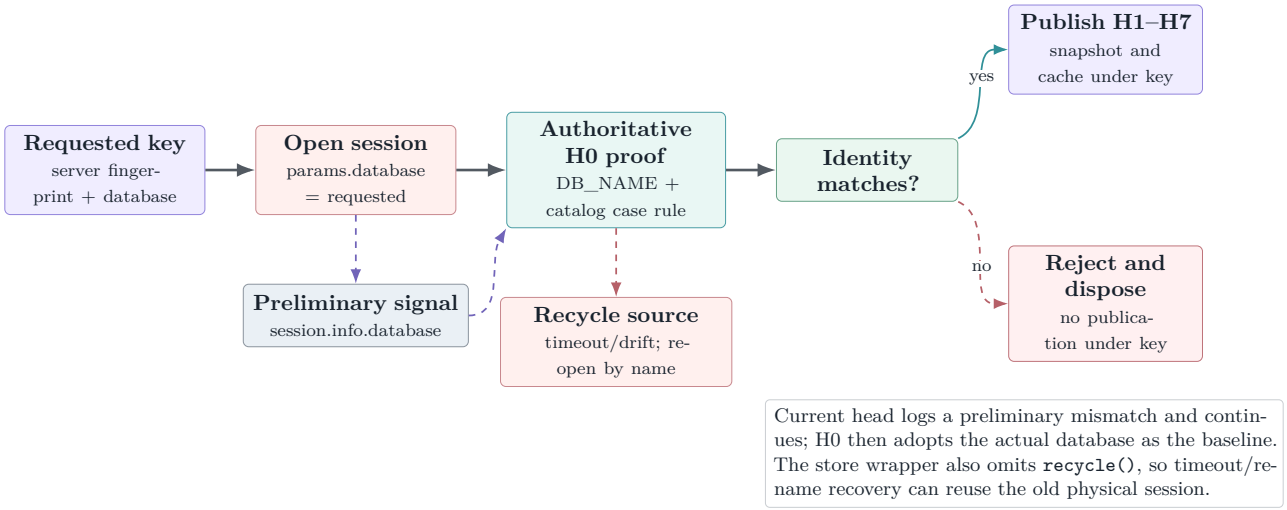



Figure 4: Database identity must be a publication gate. Detection without rejection is observability, not key correctness.

**Current-head review finding**

Two local changes are necessary but not sufficient: forward `recycle: () => inner.recycle()` through the wrapper, and fail closed when the authoritative H0 identity differs from the requested key under the catalog's comparison rule. Tests should cover a backend that ignores `params.database`, case-insensitive spelling variation, a true case-sensitive mismatch, rename drift, and watchdog recovery proving that the next operation uses a new session.

## 6 Catalog model: compact truth with explicit unknowns

### 6.1 Structure of arrays and immutable publication

`CatalogBuilder` interns strings and appends parallel arrays for schemas, objects, columns, keys, foreign keys, parameters, descriptions, and exact details. An object-id index makes row ingestion constant-time per row. `CatalogSnapshot` builds read-only indexes over those arrays and exposes query methods, not mutation.

This design has three consequences:

1. repeated schema/object/type names occupy one string-table entry;
2. canonical array order is directly serializable and hashable; and
3. a published generation is internally consistent because readers never observe a partially mutated model.

The exact detail surface includes stable `column_id`, type schema/base type, system and user type IDs, UDT/assembly flags, raw max length in bytes, precision/scale, collation, default definition, exact-text

identity seed/increment, computed definition/persistence, FK constraint IDs, referential actions, column-pair ordinals, and endpoint column IDs. Defaults and computed definitions are live structural detail but are stripped before persistence.

### 6.2 Readiness is per section

Each section is one of `absent`, `loading`, `ready`, `failed`, `stale`, or `lite`. The snapshot also carries `full`, `partial`, or `lite` mode. The distinction is semantic:

- `absent`: not hydrated or intentionally excluded;
- `failed`: attempted but unavailable;
- `lite`: useful identity-level data but not the complete contract;
- `stale` in readiness: a transient rehydration state over an existing snapshot, *not* age-based staleness.

Age and validation are represented by the freshness result described in section 7. This prevents one overloaded boolean from lying about both completeness and recency.

### 6.3 Progressive hydration ladder

Table 3: The live database hydration ladder at the pinned PR head.

| Step | Section/fact | Important behavior |
|---|---|---|
| H0 | Environment | Engine edition, default schema, database collation, `CATALOG_DEFAULT` case rule, canonical `DB_NAME()`. Current fallback silently uses case-insensitive lookup when the probe fails; the safer contract is unknown/exact-only or an unavailable name-resolution section. |
| H1 | Schemas | All visible `sys.schemas`; no high-ID role-schema filter. |
| H2 | Objects | User tables, views, procedures, scalar/table functions, synonyms, CLR routines/aggregates; system-shipped rows excluded. |
| H3 | Columns/details | Visible columns only; hidden temporal/graph implementation columns excluded; exact type/default/identity/computed facts retained live. |
| H4 | PK/UQ | Primary-key and unique-constraint columns in key ordinal order. |
| H5/H5B | Foreign keys | Enabled FKs, including untrusted-but-enforced constraints; referential actions and ordered column pairs; invisible endpoints are not published as dangling edges. |
| H6 | Parameters | Routine parameters and return value ordinal 0; schema-qualified UDT display retains exact spelling. |
| H7 | Descriptions | Object/column `MS_Description` values, failure-visible and live-only under current persistence policy. |

Synonyms are honestly `lite`: identity is present, target traversal is not. The `types` section is `absent`: table-type shapes and sequences remain a downstream model enrichment. Module text is not part of H0–H7; `getModuleDefinition(objectId)` performs one safe-integer-checked, lazy `sys.sql_modules` query and caches it only for the current generation.

### 6.4 Server and auxiliary catalogs

The server service hydrates environment and database-list facts independently from database catalogs. Additional server and database folders use `AuxiliaryCatalog`: a keyed table of lazy sections with

single-flight hydration and per-section refresh. This is the extension point for security, broker, storage, programmability leaves, logins, roles, credentials, endpoints, system objects, and designer facets without bloating connect-time hydration.

Auxiliary queries use sources separate from the main hydration lane, which is the correct resource boundary. The current auxiliary engine serializes only callers for the *same* section. Different keys can issue overlapping queries through one cached STS2 session, so a shared FIFO promise lane must sit around `sessions.open()` plus `runMetadataQuery()`. Names returned by auxiliary sections remain outside diagnostic events.

### 6.5 Deterministic schema context

`buildSchemaContext()` produces bounded text from a pinned snapshot for AI or language consumers. Ordering uses an ordinal, locale-independent comparator: Unicode-default fold followed by a raw code-unit tiebreak. This avoids Electron/ICU-version drift and preserves case-distinct names. The selection algorithm precomputes FK degree, renders each candidate at most once per fidelity tier, and takes a prefix that fits the character budget; the former quadratic shrink-and-rerender path is now linear in catalog size per tier.

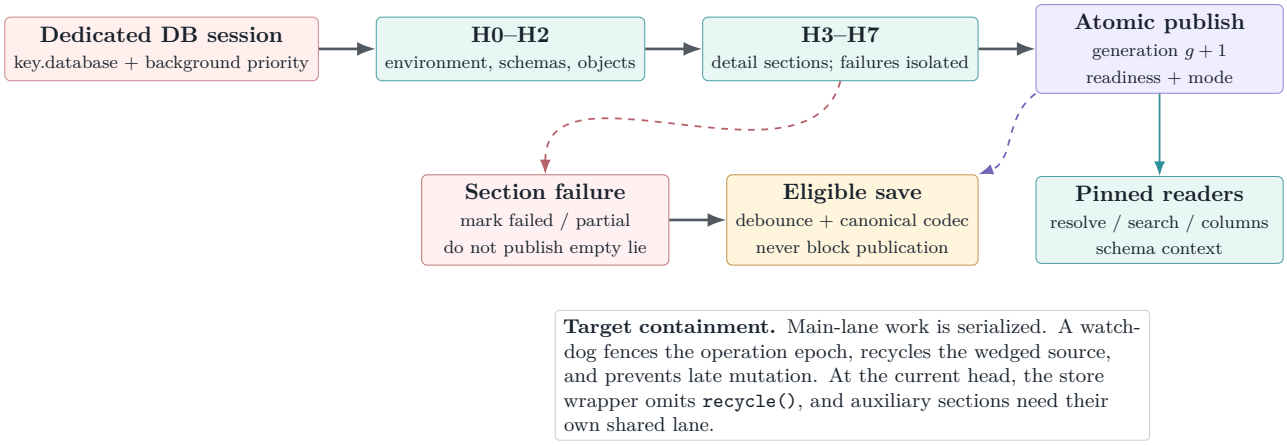



Figure 5: Rows are accumulated privately and published as one immutable generation. Per-section failure truth survives publication and persistence eligibility.

## 7 Freshness, drift, and polling

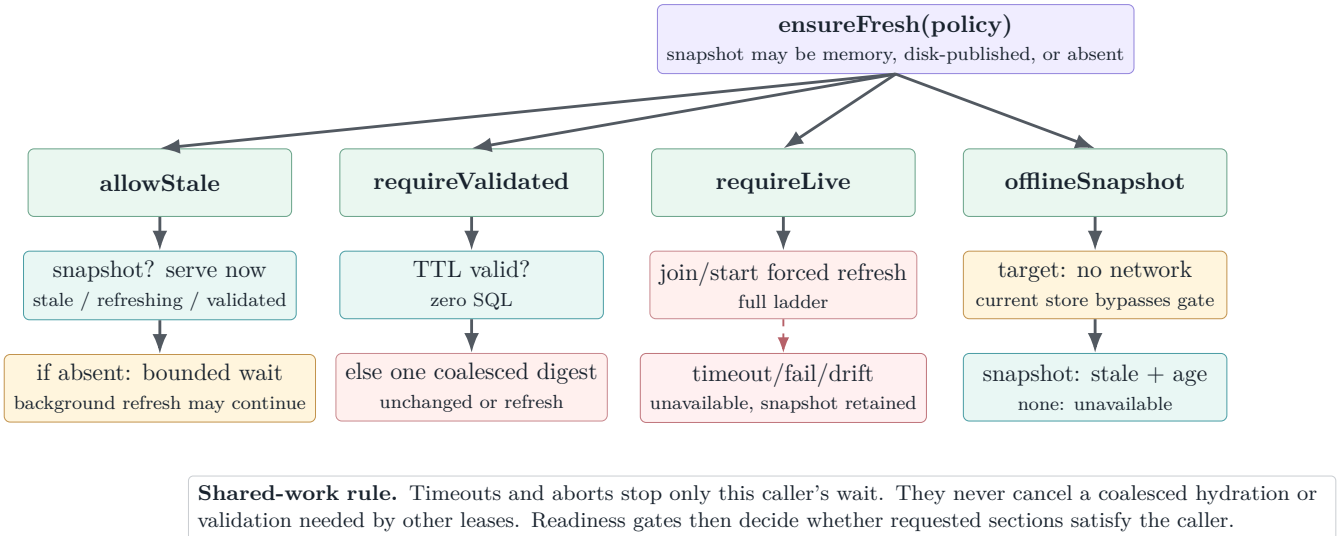
### 7.1 Consumers state a policy, not an implementation

The freshness API decouples consumer intent from cache and network mechanics. A request declares a mode, reason, optional required sections/objects, staleness or validation TTL, partial-data tolerance, disk-load permission, background-refresh preference, and a wait/abort budget.

Table 4: Freshness modes and their honest outcomes.

| Mode | Normal use | Decision semantics |
|---|---|---|
| `allowStale` | completion, AI context | Return any snapshot immediately; optionally start background refresh. With no snapshot, join hydration only up to the caller's budget. |
| `requireValidated` | diagnostics, OE browse | Accept a recent validation TTL; otherwise coalesce a cheap digest. Unchanged means validated; changed chains a full refresh. Timeout returns best snapshot as stale/unavailable, never cancels shared work. |
| `requireLive` | scripting or destructive/fidelity-sensitive work | Join/start a forced refresh. Failure, timeout, or identity drift returns `unavailable`; a retained snapshot may still ride in the result for an explicit offline choice. |
| `offlineSnapshot` | user-selected offline operation | Target: perform no backend resolution, SQL session open, poll, validation, refresh, or auxiliary query. Return disk/memory snapshot as stale with age, or unavailable if none exists. Current store-level offline wiring does not yet enforce this boundary. |

Freshness results distinguish `live`, `validated`, `refreshing`, `stale`, and `unavailable`; they also carry source, generation, capture time, stale age, wait time, validation summary, and whether background work started. Section readiness is checked *after* the freshness decision. A strict caller requiring columns cannot accept a fresh snapshot whose columns section failed.




**Shared-work rule.** Timeouts and aborts stop only this caller's wait. They never cancel a coalesced hydration or validation needed by other leases. Readiness gates then decide whether requested sections satisfy the caller.

Figure 6: The freshness decision has four modes but one invariant: return the best snapshot with an explicit truth label; never silently elevate stale or partial data.

## 7.2 Three drift paths

**Local DDL sniff.** After a successful executed batch, a leading-keyword lexer recognizes `CREATE`, `ALTER`, `DROP`, and `SP_RENAME`; `EXEC`/`EXECUTE` trigger a conservative digest check. A burst while hydration is active collapses to at most one follow-up refresh.

**Cheap digest.** The T1 query hashes visible object count plus object ID, schema ID, byte-exact name, and modify date. Including schema and binary name fixes the classic `sp_rename`/`ALTER SCHEMA TRANSFER`

blind spot. It remains evidence of likely equality, not proof: column/key/FK/parameter-only drift awaits future section/object digests or an explicit refresh.

**Explicit policy.** A strict consumer can force validation or a full live refresh. This is the correctness backstop when polling is disabled, suspended, or intentionally sparse.

### 7.3 Poll governance

The default base interval is 60 seconds with multipliers $1, 2, 5$ and independent $\pm 10\%$ jitter. Consecutive unchanged verdicts back off; refresh or detected activity resets to base. Validation fan-out is capped at two per server fingerprint, so one sleeping server cannot starve another and a resume event cannot open dozens of digest queries against one server.

After two minutes unfocused, polling suspends without SQL. Refocus schedules immediate, limiter-bounded validation and re-jitters timers. Azure SQL Database and Synapse serverless engine editions (5 and 11) poll at the cap so a background digest is less likely to defeat auto-pause; consumer activity still invokes TTL validation. Setting `mssql.metadata.pollSeconds=0` disables polling.

## 8 Persistent cache design

### 8.1 Purpose, defaults, and trust boundary

The cache reduces first-use latency and supports explicit offline browsing; it is not the source of authority. It is off by default, stored under the extension's local or remote `globalStorage/metadata-cache/v1`, limited by default to 14 days, 256 MiB total, and 32 MiB compressed per entry, with a five-second save debounce.

A cache entry is attacker-shaped or corruption-shaped local input. The read path must therefore prove four independent properties before publication:

1. **key ownership**: the manifest belongs to the requested server/database key;
2. **physical integrity**: referenced compressed bytes exist, fit bounds, and match SHA-256;
3. **logical integrity**: decoded canonical content matches the manifest's content hash and structural invariants;
4. **policy compatibility**: privacy/readiness is intersected with current settings rather than elevated.

The current implementation is strong on physical bounds, gzip limits, shape/ownership validation inside the payload, content-addressed filenames, and policy intersection. It does not yet bind the manifest to the requested key or recompute the canonical `contentHash` after decode.

### 8.2 Canonical payload and manifest

The `cm2` payload is canonical-order JSON over environment, interned strings, and structure-of-arrays fields, gzip-compressed. A model-version change is a clean miss rather than an in-place migration. Two hashes serve different jobs:

- manifest `payload.sha256`: full SHA-256 of the compressed payload file;
- `contentHash`: `csh_` plus a truncated base64url SHA-256 of canonical uncompressed JSON, intended to identify logical equality across processes/compressions.

The manifest records producer versions, writer ID, server fingerprint, database hash and optional exact database name, capture/generation/source, validation summary, environment, readiness, mode, counts/-sizes, privacy policy, and the content-addressed payload filename.

**Manifest key binding**

`readEntry(key)` structurally validates the manifest but does not compare `manifest.key.serverFingerprint`, `databaseHash`, or `databaseExact` with the requested key. A valid entry copied into another hashed directory can therefore be served under the second key. Centralize a `manifestMatchesKey()` check and use it in both manifest-only and full-entry reads; treat mismatch as a distinct safe miss/quarantine reason.

## 8.3 Privacy intersection

Current declared settings force description and module-definition persistence off. The codec strips default/computed SQL definitions before writing, and row counts are absent. The disk payload therefore contains structural catalog facts: schema/object/column/type/key/FK/parameter identities and exact non-definition facets.

Load applies policy intersection in both directions. A section disallowed by current policy, absent from payload, or never supported in v1 becomes `absent`; it never becomes ready-but-empty. A policy-ID mismatch is a normal intersected hit, not corruption.

**Validity warning**

"Hashed paths" are not encryption. The manifest may contain the exact database name, and the payload contains structural names and types. This is sanctioned local metadata, potentially sensitive, under VS Code global storage. The contract is exclusion, integrity, boundedness, and safe diagnostics—not confidentiality from a user or process that can read the directory.

## 8.4 Read protocol

A reader begins with the manifest because it is the trust root. It validates format/codec/model/shape, resolves the content-addressed payload, checks actual compressed size, verifies SHA-256, enforces a hard decompressed bound during gunzip, parses JSON, validates parallel-array and referential invariants, intersects policy, adopts arrays, and preserves the original capture time/generation.

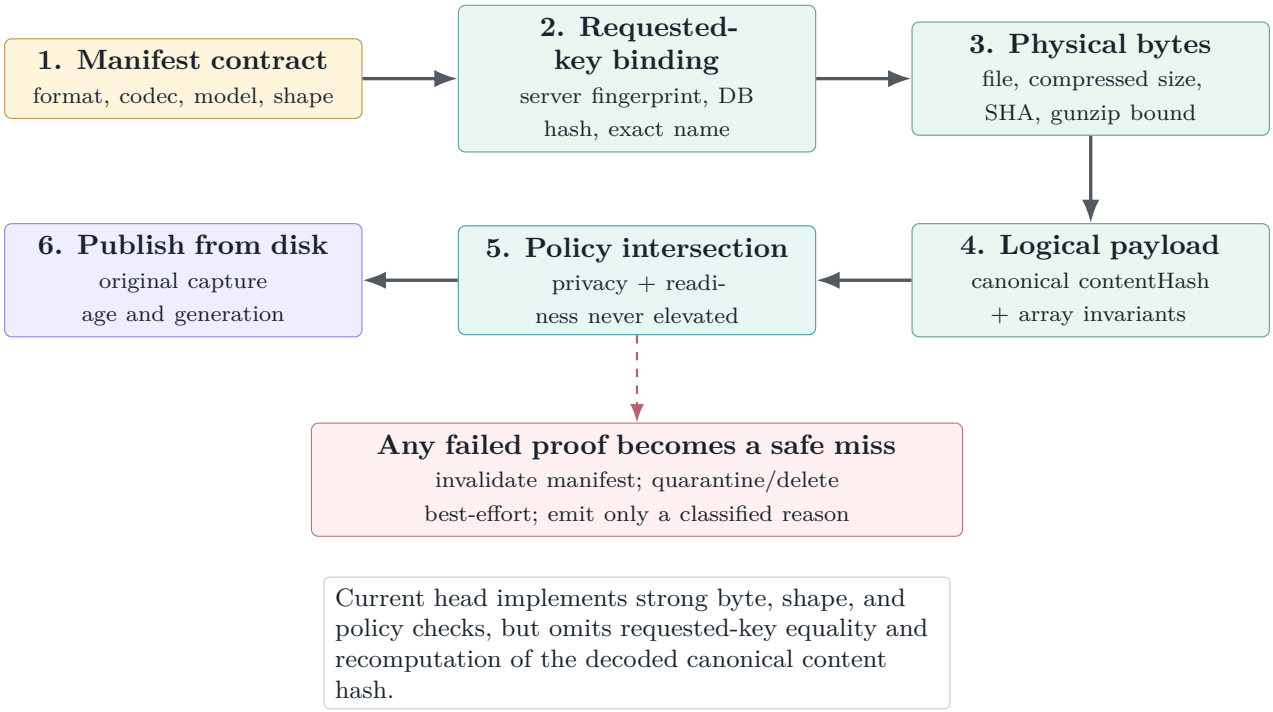



Figure 7: Complete cache trust chain. A structurally valid manifest is not trusted merely because it sits in the expected directory.

## 8.5 Write protocol and cross-window authority

Payloads are content-addressed. A writer serializes and compresses, checks the cap, writes a same-directory temporary file with fsync, renames the payload into place, then writes and renames the manifest last. Transient Windows rename failures retry with bounded jitter. Content addressing prevents a manifest from accidentally referring to another writer's bytes with the same fixed filename.

It does not, by itself, decide which capture is authoritative. The current coordinator reads the manifest, compares wall-clock capture plus a process-local generation, then later writes a new manifest. The comparison and publication are not one critical section, and generations from different extension hosts/windows are not globally ordered.

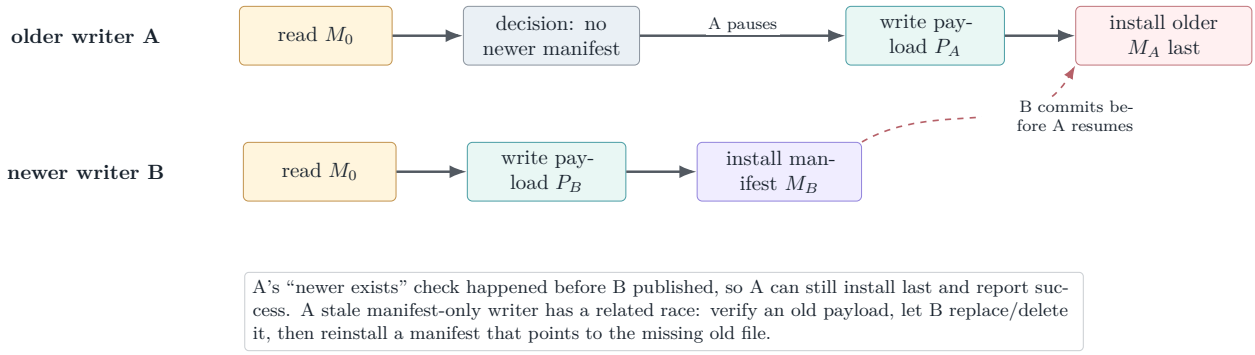



Figure 8: Uncovered two-window interleaving. Content-addressed payloads prevent byte mixing but do not make the authority decision atomic or globally ordered.

Table 5: Required cache publication properties.

| Property | Current mechanism | Required repair |
|---|---|---|
| Payload/manifest byte coherence | Content-addressed payload + manifest-last + SHA | Strong; retain bounded manifest re-read for reader/writer races. |
| Newer capture cannot be overwritten | Pre-read wall clock plus local generation | Use a per-key cross-process lock around re-read/decision/manifest commit/cleanup, or immutable candidate manifests with deterministic reader selection. Do not compare process-local generations as global time. |
| Manifest-only update cannot resurrect stale pointer | Check old payload existence, then rewrite later | Include the decision and commit under the same per-key authority mechanism; verify the referenced payload at commit. |
| Logical equality is trustworthy | Manifest `contentHash` is used for skip-save | Recompute canonical content hash after decode and compare before adoption/manifest-only equality decisions. |
| Reader key isolation | Directory path implies ownership | Compare manifest key fields to the requested key explicitly. |

### 8.6 Eviction, clearing, and index status

Maintenance coalesces re-entrant calls and throttles filesystem operations. Removal order is corrupt/unsupported and aged entries first, then least-recently-used entries until bytes fit. Manifest-less payloads, old temporary files, quarantine files, and superseded payloads receive an orphan grace. Clears remove manifests first, making remaining bytes inert before deleting payloads.

The JSON index is a convenience cache for listing/access/LRU rather than the trust root. Because each window caches and rewrites the whole index, concurrent access updates can be lost. Manifests allow rebuild, so this is lower severity than catalog corruption, but the design should either document approximate cross-window LRU or merge the latest index under the same locking strategy used for authoritative publication.

## 9 Language-service fit

### 9.1 The contract supplied by this PR

The future native language engine needs stable environment and catalog views, not direct knowledge of cache files, SQL sessions, or mutable hydration. The metadata substrate supplies the raw ingredients:

- environment: current/default database/schema, catalog case behavior, engine edition, collation;
- name resolution: exact/default-schema/ambiguous/not-found/section-unavailable;
- schemas, objects, columns, keys, FKs, parameters, descriptions, search and system/auxiliary projections;
- a generation that invalidates per-request derived caches; and
- explicit readiness/freshness so diagnostics can suppress instead of hallucinating.

The language-service design in the notes adds a thin provider interface and an `IPinnedMetadataView`. That provider should adapt `CatalogSnapshot` without exposing implementation classes. Every feature request pins once; all synchronous completion/binding operations read that generation only. Lazy module definitions remain the exceptional asynchronous resolve path.

### 9.2 No network I/O on the keystroke path

Completion, hover, signature help, and diagnostics should never await H0–H7 while the editor is responding. The intended pattern is:

1. acquire/retain a database lease when a document/connection binding becomes active;
2. call an appropriate freshness policy outside or at the bounded edge of interactive work;
3. pin one snapshot for the feature request;
4. answer synchronously from it, including local overlay symbols; and
5. request hydration/refresh asynchronously when data is absent or stale.

Completion can use `allowStale` and surface available candidates immediately. Diagnostics use `requireValidated` with a short budget and suppress schema claims if columns/objects are unavailable. Scripting uses `requireLive` because emitting destructive or fidelity-sensitive DDL from stale metadata is unacceptable.

Table 6: How planned language features consume metadata honesty.

| Feature | Suggested policy | Required sections | Degradation |
|---|---|---|---|
| Completion | `allowStale` | objects; columns when member access | Offer known candidates; omit unavailable detail; refresh in background. |
| Hover/signature | allow stale or validated | objects, columns, parameters | Show identity/type that is known; do not fabricate docs or parameter lists. |
| Diagnostics | `requireValidated` | objects + columns | Suppress schema diagnostics on timeout/-partial metadata; syntax diagnostics remain independent. |
| Definition | validated + lazy resolve | objects; module definition on resolve | Navigate to stable synthetic anchor; fetch module text only on explicit resolution. |
| Scripting | `requireLive` | all fidelity-critical sections | Refuse or offer explicit offline/stale path with fidelity note. |
| AI context | `allowStale` | bounded projection | Use deterministic, size-bounded context; include readiness, never prompt with silent gaps. |

**Scope boundary**

PR #22836 does not instantiate this language provider or switch the existing SqlToolsService language client. Its only language-client change is to gate STS2 launch arguments through the private-preview predicate. Language correctness therefore belongs to the following consumer PR, while catalog/provider correctness belongs here.

## 10 STS2 integration and endpoint evolution

### 10.1 What STS2 actually exposes today

At `upstream/dev/query` commit `fe8300939`, STS2 shares stdio with the legacy service. The multiplexer routes `v2/*` methods to STS2 and leaves other traffic on the legacy route. The generated v2 contract contains initialize, diagnostics, and fatal-notification methods; connection open/cancel/close methods; and query execute, result-set, rows, message, complete, acknowledgement, cancel, and dispose methods.

There is no `v2/metadata.*` method in that contract. The vscode-mssql capability registry correspondingly reports:

- `metadata.catalogSql`: supported;
- `metadata.endpoints`: unsupported with reason `notImplemented`.

The metadata engine therefore opens dedicated STS2 connections and sends fixed catalog SQL through `v2/query.execute`. `runMetadataQuery()` collects compact row pages, treats server error messages as failure, waits for the matching query handle completion, and only then starts the next H-step. The STS2 adapter owns query IDs, page acknowledgements, terminality, and disposal; the metadata layer sees the data-plane session abstraction.

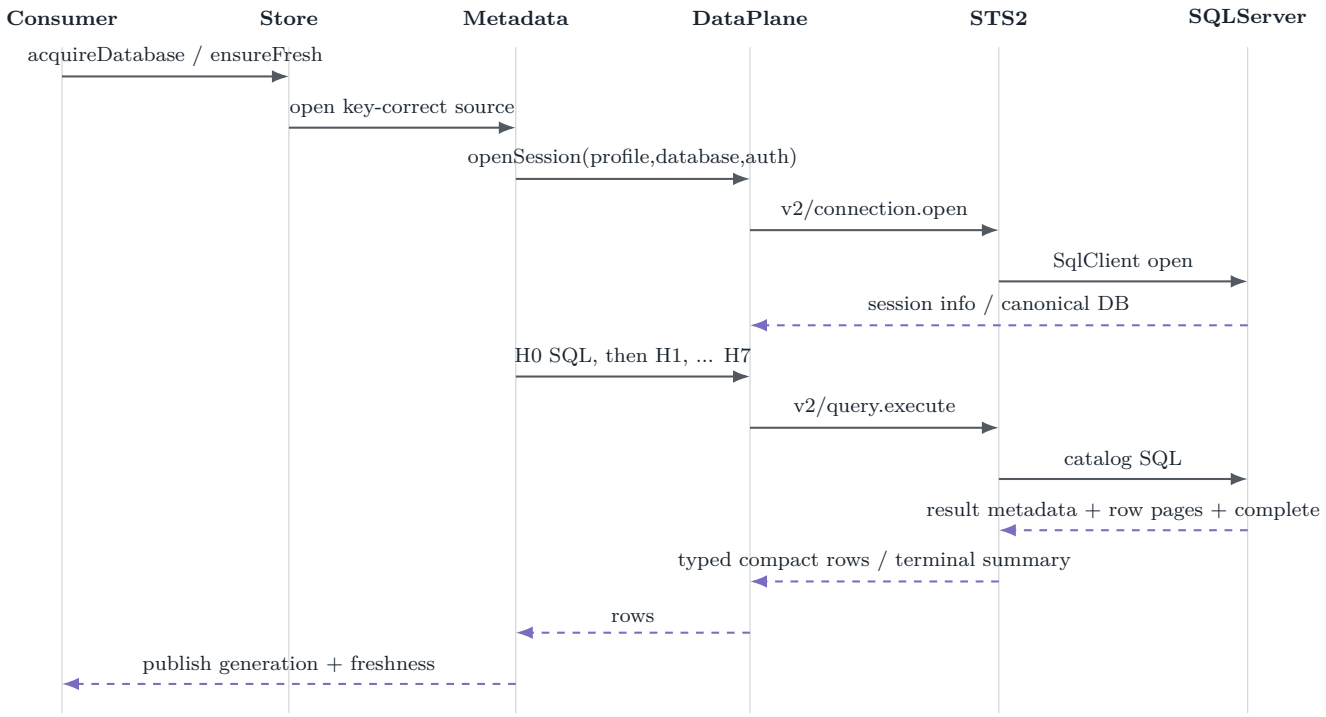



Figure 9: Current end-to-end path. The metadata service is a catalog-aware client over generic STS2 query execution, not a server-side metadata RPC.

## 10.2 Why dedicated sessions are the right bridge

STS2 permits one active query per connection and enforces page-window backpressure and exactly one terminal completion. The database engine correctly serializes its H-ladder, digest, and lazy module work on one session. User query sessions remain independent, so main-lane catalog work does not race with F5 execution.

The auxiliary boundary is only partially complete: server/database auxiliary catalogs have separate sources from the main lane, but different auxiliary section keys can overlap through the same source's cached session. The executable STS2 scenario `second-query-while-active-busy.yaml` expects the second execute to fail with `Sts2.Busy`. A per-catalog FIFO lane is therefore part of the adapter contract, not optional optimization.

### 10.3 One owner for serverless wake and retry

The SQL Data Plane has two public entry shapes: top-level `SqlDataPlaneService.openSession()` and a service view returned by `service()` or `serviceForProfile()`. The current top-level method obtains the public view and wraps `view.openSession()` in `openWithServerlessWake()`; the view's `openSession()` also wraps the raw provider call in the same helper. The resulting call graph is outer wake → inner wake → backend open.

Nested wake policy can multiply ARM status checks, delay windows, and open attempts, and makes attribution ambiguous. Choose one owner. The simplest option is for the public view to own requirements, provider availability, session registration, and wake, while the top-level method delegates directly after its pure capability/fallback decision. A spy-based unit test should assert one wake invocation and the exact backend open-attempt count for paused, resuming, and ordinary profiles.

**Related open SQL Data Plane thread**

A current Copilot thread also notes that a failed provider `canOpen()` result is routed through the capability-unsupported error helper, even when the failure represents availability/handshake state. Keep that classification fix separate from the duplicate-wake repair: capability mismatch, backend unavailability, and serverless resume are different error/retry domains.

### 10.4 How dedicated metadata endpoints can fit later

A future STS2 contract may add endpoints that return server/database sections directly, potentially with server-side paging, digests, or change tokens. The clean migration point is the `MetadataSessionSource`/data-source adapter:

1. keep `CatalogSnapshot`, store keys, lease lifecycle, freshness policies, persistence, and consumer provider unchanged;
2. add a source that checks `metadata.endpoints` and calls the new methods;
3. retain catalog-SQL fallback only where its declared fidelity is sufficient; and
4. prove endpoint-vs-SQL provider equivalence with the same snapshot fixtures and invariant suite.

The endpoint should not smuggle consumer policy into STS2. The server can own efficient catalog acquisition and stable wire shapes; the extension should continue to own cross-consumer leases, local persistence policy, VS Code focus/offline behavior, and language/UX degradation.

## 11 Logging, diagnostics, and privacy

### 11.1 Extension-side event vocabulary

The feature emits through the unified `diag` core, where every field is classified before reaching live-tail, session storage, perftest, or self-test sinks. Principal event families are:

Table 7: Metadata observability surface at the extension layer.

| Family | Representative events | Safe diagnostic meaning |
|---|---|---|
| Engine | `metadata.hydrate`, `metadata.ensureFresh`, and validation/drift events | Duration, generation, policy mode/reason, freshness/source, wait, error class; database fields are classified paths. |
| Store | acquire server/database, dispose lease, server/aux hydration, access drift, key-correctness tripwires | Short fingerprint, key kind, cache hit/miss, refcount, generation/readiness/counts; no catalog item names. |
| Cache | `load`, `save`, `hit`, `miss`, `corrupt`, `evict`, `clear`, and `raceLost` | Fingerprint/hash prefixes, byte/duration counts, fixed age buckets, safe enums, policy-intersection boolean. |
| Performance | `mssql.metadata.cache.warmAcquire.begin` / `mssql.metadata.cache.warmAcquire.end` | Same-process monotonic marker pair; object count and wait time; feature bucket `metadata`. |

The generated observability registry is the source of truth, mirrored into the embedded perftest package. Conformance tests check that source-emitted names are registered and marker pairs agree. `Perf.featureFor()` maps `metadata.*`, `metadataStore.*`, `metadataCache.*`, and `mssql.metadata.*` to the metadata feature.

### 11.2 Allowlist and forbidden fields

Allowed event attributes include short non-reversible fingerprint/hash prefixes, generations, readiness summaries, fixed stale-age buckets, byte/duration/wait counts, safe enum results/reasons/tiers, booleans, and error classes. Forbidden fields include raw object/database names unless classified and redacted at the common choke point, SQL text, result rows, endpoints, connection strings, secrets/tokens, prompt text, module definitions, description values, and content-hash prefixes pending policy review.

The key-correctness tripwires intentionally classify expected/actual database values as `source.path`; they do not bypass the central redactor. These events prove that a mismatch was detected, not that publication was prevented. Cache status/clear UI may display local database names to the user, but those names do not become telemetry.

### 11.3 Extension diagnostics versus STS2 envelopes

These are complementary layers:

- **Extension diagnostics** explain consumer/store/cache decisions: why a snapshot was served, rejected, refreshed, or evicted.
- **STS2 envelopes** explain transport/runtime causality: RPC input, reducer output, effects, query pages, terminal events, configuration changes, and runtime health.

STS2 journals each sanitized envelope before dispatch, then publishes it in sequence order to best-effort metrics/live-tail/custom sinks. Product capture replaces SQL text and row cells with digest wrappers; secret material lives only in an in-memory side table behind opaque references. The envelope carries run ID, gapless sequence, UTC timestamp, kind/type, session/correlation, causal sequence, config version, payload digest, optional payload, and redaction metadata.

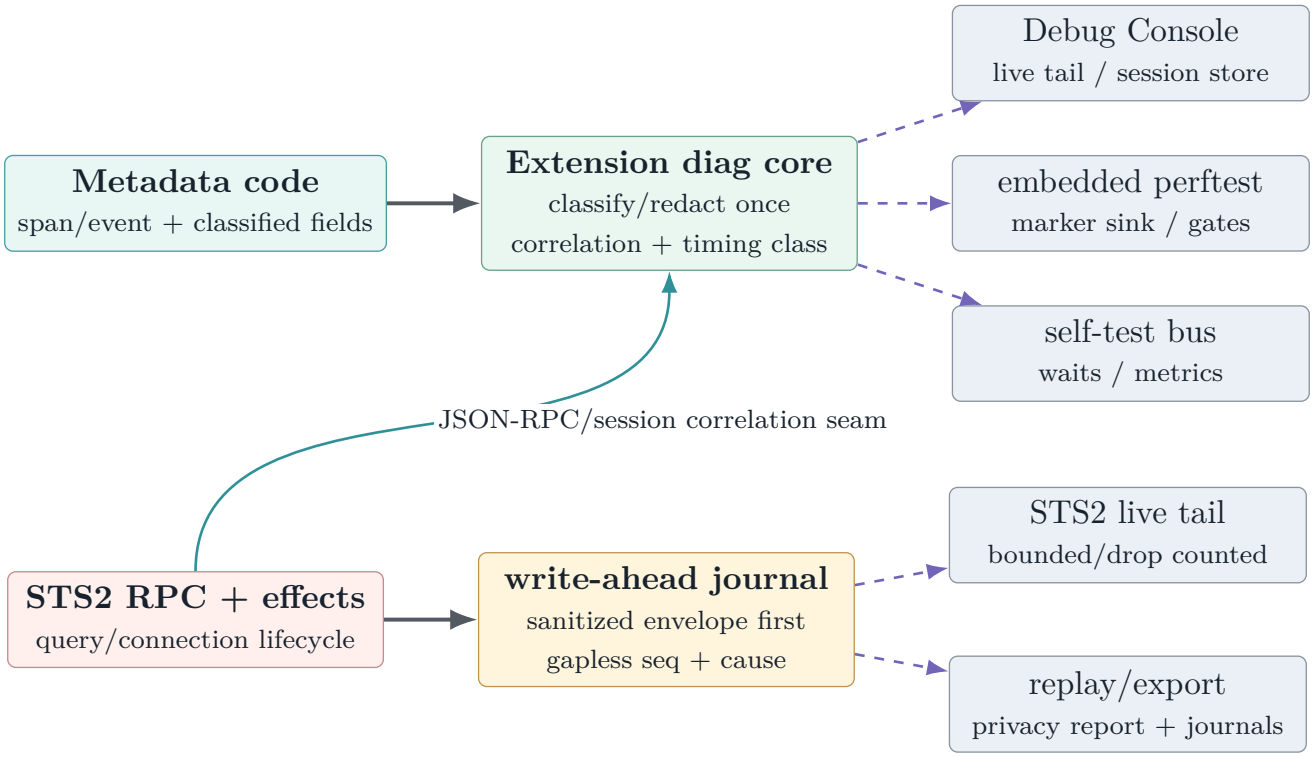



Figure 10: Two observability planes. Extension events explain metadata policy; STS2 envelopes explain transport causality. Both are bounded and privacy-classified, and can be stitched without copying SQL or rows into product telemetry.

# 12 Test design and coverage

## 12.1 Strength of the current test strategy

The focused suite is unusually good at semantic failure modes: immutable publication, section readiness, deterministic ordering, exact catalog detail, same-key acquisition, lifecycle disposal, codec bounds, malformed payloads, torn writes, privacy canaries, poll governance, and large-catalog scaling. Prior review comments also drove concrete fixes for quadratic projection, delimiter collisions, UDT rendering, hidden columns, role-owned schemas, untrusted FKs, decompression bounds, content-addressed payloads, and shutdown flush.

The remaining problem is not a lack of test volume; it is branch selection. Several current tests prove a nearby property while avoiding the unsafe interleaving or policy boundary.

## 12.2 Current focused suite inventory

The current CI log at the pinned head reports 151 passing tests across the eleven new metadata/cache/-catalog files below. Adding the touched profile-identity, private-preview/manifest, and SQL Data Plane registry files yields 197 passing tests in the immediately affected suite set; that second number is not a claim that every assertion in pre-existing touched files was introduced by this PR.

Table 8: Metadata test inventory at `9955ff9e6`.

| Suite | CI count | Principal invariants |
|---|---|---|
| `metadataCatalog` | 35 | H-ladder parsing, readiness/partial failures, resolution/search, DDL keyword detection, descriptions, lazy definitions, burst coalescing, hidden columns, safe IDs. |
| `metadataCatalogDetail` | 6 | Exact column/FK facts, UDT spelling/schema, identity text, defaults/computed detail, FK actions, and column identity. |
| `metadataDeterminism` | 6 | Ordinal ordering, byte-stable schema context, catalog-collation case behavior, and digest-v2 rename detection. |
| `metadataFreshness` | 8 | Four policy modes, TTL, timeout-as-race, section gates, and stale/live/unavailable honesty. |
| `metadataPoll Governance` | 12 | Backoff, jitter bounds, focus suspension/resume, serverless cap, limiter behavior, and failure retry. |
| `metadataStore` | 13 | A/B key isolation, same-key single flight, routed sessions, leases, refcounts, idle TTL/LRU, cleanup, and server catalogs. |
| `metadataStoreCache` | 6 | Disk-before-live publication, manifest generation/digest baseline, background refresh, offline seeded hit, recovery, and A/B isolation with cache. |
| `metadataStoreScale` | 6 | 10k-object hydration/counts, ordering/search, 1k-column table, bounded schema context, and warm reacquire without rehydration. |
| `metadataStoreService` | 3 | Default-off composition, enabled coordinator, and data-plane-disabled fail-fast without backend acquisition. |
| `metadataCacheCodec` | 20 | Live/disk round-trip equivalence, byte stability, content-hash perturbation, privacy stripping, strict shape/range/ownership validation, and model/version misses. |
| `metadataCacheStore` | 36 | Path containment, byte integrity, oversize bounds, quarantine, rename retry, torn writes, selected two-writer/read races, eligibility/skips, policy intersection, manifest-only save, privacy canaries, index/eviction/clear/settings. |
| **Metadata subtotal** | **151** | Current GitHub Actions per-file counts at the pinned head; zero failures. |

This inventory demonstrates broad mechanism coverage, but its wording matters. For example, "A/B isolation" currently proves ordinary key/path separation; it does not prove that a structurally valid A manifest copied into B's already-derived directory is rejected. The regression matrix below names those stronger boundary tests explicitly.

Table 9: Regression matrix required to close the current review findings.

| ID | Invariant | Current coverage | Missing deterministic proof |
|---|---|---|---|
| F1 | Offline setting means zero live work | Guaranteed disk hit, `pollSeconds=0`, session count | Miss; default polling; resolver count; explicit refresh/strict freshness; auxiliary; online→offline. |
| F2 | Wrong database never publishes; drift/timeout reopens | Correct-routing fixture and later digest drift | Backend ignores database; authoritative H0 mismatch; CS names; wrapper recycle; new session after drift/watchdog. |
| F3 | One active query per auxiliary session | Same-section single flight | Two different section keys overlap; fake rejects concurrency; both complete with max active 1. |
| F4 | Cache entry belongs to requested key | Path derivation/containment | Copy valid A manifest+payload into B directory; B must miss/quarantine. |
| F5 | Newer authority cannot be overwritten | Winner observed between one writer's rename and reread | Older writer installs last; newer capture has lower local generation; manifest-only/full interleave; equal timestamps; clock skew. |
| F6 | One serverless wake policy per open | Registry/service-view tripwires | Spy exact wake/status/open attempts through top-level and direct-view paths. |
| F7 | Wait abort listener is removed | Server service has listener test | Database service success, timeout, and repeated reused-signal paths. |
| F8 | Unknown case rule never folds optimistically | H0 success and collation divergence | H0 failure on case-sensitive names differing only by case. |
| F9 | Server refresh recovers from hung completion | Caller timeout race | Hung completion, later refresh, recycled source, late result fenced. |
| F10 | Manifest logical hash matches decoded payload | SHA and canonical hash unit tests separately | Valid compressed SHA/payload shape with incorrect manifest `contentHash`. |

### 12.3 Persistence tests remain protocol tests

Cache tests should continue to model filesystem operations as a protocol, not as ordinary serialization coverage. The completed suite already covers many crash points and corruption classes. The next additions must distinguish three layers:

- **byte safety**: size, SHA, gzip bound, JSON and array invariants;
- **identity safety**: requested key and logical content hash;
- **authority safety**: one atomic, globally comparable publication decision per key.

A result such as `raceLost` is only meaningful if the final manifest is known to represent the winning authority. Post-save writer-ID comparison alone does not prove that when an older writer installs last.

### 12.4 Embedded perftest coverage

The embedded scenario `metadatacache-warm-acquire` is the correct performance boundary: cold fill/save, a fresh store in offline mode, disk verification and rehydration/publication, source-honesty assertion, and begin/end markers around only the warm acquire. It is exploratory rather than an official SLA gate. Before graduation it should run across representative catalog sizes, Windows/macOS/Linux, local and remote extension hosts, and repeated cold/warm profiles with pre-registered p50/p95 budgets.

The current offline defect is directly relevant to this scenario: a benchmark can report a disk snapshot while still resolving live infrastructure or arming later polling. The probe should assert resolver calls and session opens are zero, not only that the returned source label is `disk`.

Local package verification in this work session built the embedded perftest workspace and ran 347 tests passing with 16 explicit skips. Fifteen skips are central-store integrations and one is vendor-sync; the Query Studio scenario tests (158) and VS Code spawn tests (2) passed. This verifies registry/contracts/harness behavior, not a live SQL warm-cache benchmark run against a provisioned `PerfHarness` database.

### 12.5 Coverage evidence and its limit

Codecov at the pinned head reports 91.09384% patch coverage with 894 changed lines uncovered. Project coverage rises from 77.17% at the PR base to 77.96% at the head. Core metadata files are strong—catalog model 97.22%, codec 95.81%, coordinator 96.20%, store 94.47%, and engine 93.44%—while host wiring and broad auxiliary surfaces are weaker: main controller 9.38%, database auxiliary sections 73.75%, auxiliary engine 77.44%, and store composition service 86.53%.

Coverage supports the focused suites but does not settle the review findings. Race ordering, dynamic network authorization, key binding, and retry ownership are semantic properties; they require the deterministic adversarial tests in table 9, not merely more executed lines.

## 13 Build and validation record

### 13.1 Current GitHub state observed independently

At `9955ff9e6`, analysis/CodeQL, Azure PR validation, packaging, normalization, CLA, and all six platform VSIX-launch jobs pass. The GitHub `build-and-test` check is red, but its metadata-relevant facts are more specific:

- extension install/build, lint, package, and MSSQL unit-test steps pass;
- the GitHub unit runner reports 3,975 passing, 0 failing, and 12 skipped;
- every metadata file in table 8 reports zero failures;
- the aggregate job fails because SQL Projects unit tests return exit code 1 and one of 17 VSIX smoke tests—"MSSQL add connection webview is loaded correctly"—fails.

This is not a green-PR claim, and it is not evidence of a metadata failure. It is a current whole-job status that should be resolved or rerun before merge.

### 13.2 Independent local validation

The local workspace was updated and built at the pinned commits before this report. Results are kept separate from PR-author comments and GitHub CI:

Table 10: Independent local validation snapshot, 28 August 2026.

| Lane | Result | Interpretation / explicit exclusions |
|---|---|---|
| vscode-mssql build | Pass | Full extension build completed on `9955ff9e6`. |
| vscode-mssql full tests | 5,880 pass; 13 fail; 31 skip | All metadata-specific tests pass (176 metadata/-cache/catalog assertions in the local selection). The 13 failures cluster in Schema Visualizer, Query Studio bundling/activation, pinned results, OE v2, feature capture, Dockerode, and a missing PLP plugin—outside this PR's metadata files. |
| Embedded perftest | 347 pass; 16 skip | All package tests/builds pass; skips are described above; the live provisioned warm-cache scenario was not executed. |
| sqltoolsservice Release solution | Pass; serial MSBuild with shared compilation disabled | 0 errors; two ScriptDom locale warnings. |
| STS2 unit + E2E | 496 pass / 10 SQL-server-gated skip; 6/6 E2E | Deterministic/runtime contract and process interop green; engine-gated skips explicitly reported. |
| STS auth / language | 9/9; 76/76 | Adjacent authentication and current language-service lanes green. |
| ServiceLayer unit | 1,944 pass; 1 fail | One platform/provider DNS failure, not metadata logic. No on-prem integration lane was provisioned. |

**Evidence boundary**

The local runs establish that the pinned trees build and that existing metadata, cache, STS2, and embedded-perftest suites pass. They do not close F1–F11: the required adversarial branches and interleavings are absent from the current tests, and the live provisioned warm-cache and on-prem SQL matrices were not run.

### 13.3 Author- and review-reported evidence

The PR body reports additional focused host-composition, data-plane capability, observability, localization, packaging, and full-suite results. Prior review threads report targeted performance probes and repeated focused suites after earlier fixes. These are useful confidence signals and should remain in the PR record, but they are **Reported evidence** rather than rerun as part of this review pass.

Two current review threads were unresolved at the pinned head and should receive explicit dispositions rather than duplicate comments:

- provider unavailability from a public SQL Data Plane view is classified as `capabilityUnsupported`;
- the cache-entry QuickPick combines a fingerprint prefix with localized text in one `description` string.

## 14 Performance mechanisms and evidence quality

The implementation removes the most important algorithmic traps:

- constant-time object-ID lookup during build;
- value-by-value row-page append rather than spreading unbounded arguments;
- snapshot indexes for name/prefix lookup;
- linear schema-context candidate rendering per fidelity tier;
- same-key acquisition, validation, and forced-refresh coalescing;
- content-equal cache saves reduced to a manifest update;
- delayed/throttled maintenance outside activation;
- focus-gated, backoff/jittered, per-server-limited polling.

Prior review measurements show that the fixed scale paths are orders of magnitude better than the earlier quadratic implementation. Those measurements are review-machine observations, not product SLAs. The more urgent performance risk at the current head is policy multiplication: nested serverless wake can multiply open/status/retry work, and un-serialized auxiliary sections convert ordinary concurrent expands into failed/retried work.

A graduation performance plan should separate:

1. live H0–H7 hydration CPU, SQL time, memory peak, and payload size;
2. warm cache read, hash/decompress/validate, and publication time;
3. interactive snapshot query/projection latency with no network;
4. background poll/validation cost under focus, resume storms, and serverless pause; and
5. multi-window contention/lock hold time after the authority fix.

# 15 Mechanism-to-evidence traceability

Table 11: Load-bearing mechanisms, current evidence, and remaining closure condition.

| Mechanism | Current evidence | Closure condition |
|---|---|---|
| Same-key single flight | Eight-way concurrency tests and explicit disposal coverage | Retain. |
| Immutable deterministic snapshot | Catalog/detail/determinism/round-trip/scale suites | Retain; add H0-unknown semantics. |
| Database identity | Requested database parameter, tripwire event, digest rider | Reject before publication and forward recycle; prove wrong-backend/rename/watchdog cases. |
| Offline behavior | Freshness enum and a seeded-hit test | Dynamic network policy across resolver, hydration, poll, refresh, validation, and aux. |
| STS2 one-query discipline | Main lane plus STS2 invariant/scenario | Add auxiliary FIFO and server-service watchdog/recycle. |
| Cache byte integrity | Content-addressing, SHA, gzip bound, shape validation | Add requested-key and canonical-content-hash verification. |
| Cache cross-window authority | Manifest-last, post-save reread, selected race tests | Atomic per-key authority mechanism and older-last/manifest-only interleaves. |
| Privacy at rest | Codec stripping, manifest policy, disk canaries | Retain; re-run after cache changes. |
| Data-plane capability/auth | Pure requirement checks and credential tripwires | Fix availability classification; choose one serverless wake owner. |
| Consumer seam | Lease/snapshot/freshness APIs and refactor designs | Future provider equivalence, pinned-generation, no-keystroke-I/O, and honesty corpus. |

# 16 Known boundaries and next validation gates

## 16.1 Model enrichments deliberately deferred

The substrate reports rather than conceals incomplete domains. Synonym target traversal, table-type shapes, sequences, parameter/schema descriptions, row counts, and section/object-level T2/T3 digests need new model/API shapes. They should land with consumers that can exercise them end to end, not as unreachable arrays that misleadingly say `ready`.

## 16.2 Integration work for the native language service

The following gates should precede a language cutover:

1. resolve F1–F3 so offline, identity, and STS2 lane contracts are trustworthy;
2. implement the provider adapter and prove fixture/null/catalog provider equivalence;
3. pin one generation per feature request and test a mid-request generation change;
4. prove completion/hover/diagnostics make no network call on the keystroke path;
5. run the honesty corpus: temp tables, CTEs, dynamic SQL, synonyms, cross-database names, case-sensitive catalogs, partial/offline metadata, and `USE` changes;
6. define suppression/status UX for unavailable sections and stale/offline snapshots;
7. benchmark real completion p50/p95 and document CPU/memory budgets separately from cache load.

### 16.3 Integration work for dedicated STS2 endpoints

Before flipping `metadata.endpoints` to supported:

1. add generated wire-contract methods and stable error shapes in STS2;
2. define section/readiness semantics and paging/backpressure for large catalogs;
3. preserve database-key correctness and one-terminal cleanup under cancel/dispose;
4. decide whether STS2 returns canonical arrays, row-like sections, or a provider-neutral DTO—without importing the extension cache format as wire format;
5. produce a live-SQL truth corpus and endpoint-vs-catalog-SQL snapshot-equivalence tests;
6. extend STS2 envelope/privacy canaries so metadata names and definitions follow explicit capture policy.

### 16.4 Current-head implementation plan

Table 12: Recommended repair order for PR #22836.

| Order | Repair | Why this order |
|---|---|---|
| 1 | F2 fail-closed identity + recycle forwarding | Prevent wrong-key publication and restore watchdog/rename recovery before adding more cache/live behavior. |
| 2 | F1 dynamic offline network gate | Central policy affects acquisition, engine timers, freshness, and auxiliary work; settle it before tests depend on lifecycle details. |
| 3 | F3 auxiliary FIFO + F9 server watchdog | Align every metadata session with the STS2 one-active-query and recovery contracts. |
| 4 | F4 key binding + F10 logical-hash verification | Complete cache read trust before modifying write authority. |
| 5 | F5 per-key cross-window authority | Introduce lock/candidate protocol, then rerun torn-write, eviction, and manifest-only tests. |
| 6 | F6 single wake owner + availability classification | Restore one retry/error owner and pin call counts. |
| 7 | F7 abort listener + F8 unknown case behavior + index documentation/merge | Contained hardening after the main lifecycle/protocol shapes stabilize. |
| 8 | Full focused suites, Actions rerun, real-SQL matrix, warm-acquire probe | Produce closure evidence at the new head and update this report's pins/status. |

## 17 Final assessment

**Key takeaway**

The architectural direction is good: one shared lease/store, immutable compact snapshots, explicit per-section truth, policy-routed freshness, default-off persistence, content-addressed payloads, and a future transport/provider seam are the right foundations for native language and AI-assisted SQL experiences.

The current head is not yet a safe merge point for that foundation. Several claims in the original design narrative—zero-network offline operation, key-correct publication, one-active-query auxiliary behavior, newer-writer cache authority, and one serverless wake policy—are target invariants rather

than properties enforced by `9955ff9e6`. The repair plan is bounded and testable. Once those gaps are closed, the same design can remain intact; the work is primarily to make the implementation honor its own boundaries.

**Decision statement**

Recommended disposition: retain the design, request changes on the current implementation, rerun focused and whole-PR validation at the repaired head, and update the commit pins before treating this PDF as a merge artifact. The companion code-review Markdown contains exact line anchors and replacement/test guidance.

## Evidence and source map

**Reproducibility note**

This report is pinned because the PR and related branches are active. Review threads, Actions conclusions, and code locations are observations at the stated snapshot, not timeless properties. Re-fetch the head, rerun the specified regression and whole-PR lanes, and update the title-page pins and disposition before using the PDF after any new commit.

| Evidence class | Primary sources used |
|---|---|
| PR state | PR #22836, current body/discussions/check metadata, head `9955ff9e6`, base `69f54be2a`; all 46 changed filenames. |
| Core implementation | `extensions/mssql/src/services/metadata/catalogModel.ts`; `metadataService.ts`; `metadataStore.ts`; `metadataStoreService.ts`; `serverMetadataService.ts`; `auxiliaryCatalog*.ts`; profile/auth/fingerprint modules. |
| Cache implementation | `extensions/mssql/src/services/metadata/cache/metadataCacheCodec.ts`; `metadataCacheManifest.ts`; `metadataCacheStore.ts`; `metadataCacheCoordinator.ts`; `metadataFreshness.ts`; `metadataCacheSettings.ts`. |
| Host/data plane | `mainController.ts`; `previewService.ts`; `sqlDataPlaneService.ts`; `languageservice/serviceclient.ts`; package settings/localization; `perfTelemetry.ts`. |
| Tests | Metadata/cache/catalog/freshness/poll/store/scale suites, profile and data-plane tests, observability conformance, large-catalog fixture, and in-memory filesystem harness; CI per-file counts plus the independent local validation in table 10. |
| Embedded perftest | `tools/perftest/packages/perftest-cli/src/scenarios/registry.ts` and the metadata warm-acquire command/marker wiring. |
| Implementation review | Companion workspace review `vscode_mssql_pr_22836_code_review.md`; every finding was rechecked against local head `9955ff9e6` and the live PR head before publication. |
| Metadata/refactor designs | `vscode-ext-refactor/metadata-docs/*`, including substrate, cache/drift, and review addendum documents; current code controls when design notes and implementation differ. |
| Language/observability designs | Native T-SQL language-service and observability notes under `vscode-ext-refactor`; treated as downstream contracts, not PR content. |
| STS2 authority | `microsoft/sqltoolsservice dev/query` at `fe8300939`: generated contract, `docs/sts2/INVARIANTS.md`, runtime/client documentation, and `test/sts2/scenarios/second-query-while-active-busy.yaml`. |